Exponentials & Logarithms

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Evaluating exponential expressions

Evaluate 2^5.

$2^5 = 32$.

2 [Foundational] • Target: Evaluating exponential expressions

Evaluate 3^{-2}.

$3^{-2} = \frac{1}{9}$.

3 [Foundational] • Target: Evaluating exponential expressions

Evaluate $16^{1/2}$.

$16^{1/2} = 4$.

4 [Foundational] • Target: Evaluating logarithms

Evaluate $\log_2 32$.

Since $2^5 = 32$, $\log_2 32 = 5$.

5 [Foundational] • Target: Evaluating logarithms

Evaluate $\log_{10} 0.01$.

Since $10^{-2} = 0.01$, $\log_{10} 0.01 = -2$.

6 [Foundational] • Target: Log laws: product and quotient

Write $\log_2 (8x)$ as a sum of logarithms.

$\log_2 (8x) = \log_2 8 + \log_2 x = 3 + \log_2 x$.

7 [Foundational] • Target: Log laws: product and quotient

Write $\log_3 \left(\frac{x}{9}\right)$ as a difference of logarithms.

$\log_3 \left(\frac{x}{9}\right) = \log_3 x - \log_3 9 = \log_3 x - 2$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Log laws: product and quotient

Combine $\log_5 5 + \log_5 25$ into a single logarithm and evaluate.

$\log_5 (5 \times 25) = \log_5 125 = 3$ (since $5^3 = 125$).

9 [Intermediate] • Target: Log laws: product and quotient

Combine $\log_2 24 - \log_2 3$ into a single logarithm and evaluate.

$\log_2 \left(\frac{24}{3}\right) = \log_2 8 = 3$.

10 [Intermediate] • Target: Log laws: the power rule

Write $\log(x^5)$ using the power rule.

$\log(x^5) = 5\log x$.

11 [Intermediate] • Target: Log laws: the power rule

Evaluate $\log_2 (4^3)$ using the power rule.

$\log_2 (4^3) = 3\log_2 4 = 3(2) = 6$.

12 [Intermediate] • Target: Log laws: the power rule

Write $2\log_3 x + \log_3 y$ as a single logarithm.

$2\log_3 x + \log_3 y = \log_3 (x^2 y)$.

13 [Intermediate] • Target: Expanding and condensing logarithmic expressions

Expand $\log \left(\frac{x^3 y}{z^2}\right)$ fully.

$3\log x + \log y - 2\log z$.

14 [Intermediate] • Target: Expanding and condensing logarithmic expressions

Condense $3\ln x - 2\ln y$ into a single logarithm.

$\ln \left(\frac{x^3}{y^2}\right)$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Expanding and condensing logarithmic expressions • Cross-Lesson: 6.01 Limits & Derivatives

Expand $\log_2 \left(\sqrt{x} \cdot y^3\right)$ fully.

$\sqrt{x} = x^{1/2}$, so the expression equals $\frac{1}{2}\log_2 x + 3\log_2 y$.

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16 [Challenging] • Target: Solving exponential equations (matching bases) • Cross-Lesson: 7.01 Integration & Applications

Solve $2^x = 32$.

$32 = 2^5$, so $x = 5$.

17 [Challenging] • Target: Solving exponential equations (matching bases) • Cross-Lesson: 9.01 Statistics & Probability

Solve $3^{x+1} = 81$.

$81 = 3^4$, so $x + 1 = 4 \Rightarrow x = 3$.

18 [Challenging] • Target: Solving exponential equations (matching bases) • Cross-Lesson: 1.01 Algebra & Equations

Solve $5^{2x} = 125$.

$125 = 5^3$, so $2x = 3 \Rightarrow x = \frac{3}{2}$.

19 [Challenging] • Target: Solving exponential equations using logarithms • Cross-Lesson: 2.01 Functions & Transformations

Solve $2^x = 10$ for x, rounded to three decimal places.

$x = \frac{\log 10}{\log 2} \approx \frac{1}{0.301} \approx 3.322$.

20 [Challenging] • Target: Solving exponential equations using logarithms • Cross-Lesson: 3.01 Sequences & Series

Solve $3^{2x} = 50$ for x, rounded to three decimal places.

$2x = \frac{\log 50}{\log 3} \approx \frac{1.699}{0.477} \approx 3.561$, so $x \approx 1.781$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Exponential growth and decay models

The graph shows an exponential function. Based on the graph, does it represent growth or decay?

a. Neither

b. **Growth ✓**

c. Cannot be determined

d. Decay

The curve rises as x increases: growth.

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22 [Foundational] • Target: Log laws: the power rule

Write $4\log x - \log y$ as a single logarithm.

a. $\log\left(\frac{x^4}{y}\right)$ ✓

b. $\log(4x - y)$

c. $4\log\left(\frac{x}{y}\right)$

d. $\log\left(\frac{4x}{y}\right)$

$4\log x = \log(x^4)$; combined: $\log\left(\frac{x^4}{y}\right)$.

23 [Foundational] • Target: Expanding and condensing logarithmic expressions

Condense $\frac{1}{2}\log x + 2\log y$ into a single logarithm.

a. $\log(x \cdot y^2)$

b. $\log\left(\frac{\sqrt{x}}{y^2}\right)$

c. $\log(\sqrt{x} \cdot y^2)$ ✓

d. $\log(2\sqrt{x} \cdot y)$

$\frac{1}{2}\log x = \log\sqrt{x}$; combined: $\log(\sqrt{x} \cdot y^2)$.

24 [Foundational] • Target: Solving exponential equations (matching bases)

Solve $4^x = \frac{1}{16}$.

a. $x = 4$

b. $x = -4$

c. $x = -2$ ✓

d. $x = 2$

$4^x = 4^{-2} \Rightarrow x = -2$.

25 [Foundational] • Target: Solving exponential equations (matching bases)

Solve $2^{2x-1} = 32$.

a. $x = 3$ ✓

b. $x = 2.5$

c. $x = 6$

d. $x = 5.5$

$2^{2x-1} = 2^5 \Rightarrow 2x - 1 = 5 \Rightarrow x = 3$.

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26 [Foundational] • Target: Natural logarithms and e

Evaluate $\ln\left(\frac{1}{e}\right)$.

a. 1

b. e

c. -1 ✓

d. 0

$\ln(e^{-1}) = -1$.

27 [Foundational] • Target: Natural logarithms and e

Evaluate $e^{\ln 7}$.

a. 7 ✓

b. $\ln 7$

c. 1

d. e^7

$e^{\ln x} = x$.

28 [Foundational] • Target: Exponential growth and decay models

The population of a town is modeled by $P(t) = 8{,}000e^{0.025t}$, where t is in years. Find $P(15)$, rounded to the nearest whole number.

a. 8,300

b. 11,000

c. 11,640 ✓

d. 12,000

$P(15) = 8{,}000e^{0.375} \approx 11{,}640$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Exponential growth and decay models

Determine whether $N(t) = 80e^{-0.015t}$ models growth or decay.

a. **Decay** ✓

b. Neither

c. Growth

d. Cannot be determined

The exponent's coefficient is negative: decay.

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30 [Intermediate] • Target: Exponential growth and decay models

Determine whether $M(t) = 30e^{0.01t}$ models growth or decay.

a. Neither

b. Cannot be determined

c. Decay

d. **Growth ✓**

The exponent's coefficient is positive: growth.

31 [Intermediate] • Target: Word problems: exponential models in context

An investment of 3,000 dollars grows according to $A(t) = 3{,}000e^{0.05t}$ dollars, where t is in years. Find the investment's value after 8 years, rounded to the nearest dollar.

a. 4,200

b. 4,800

c. 4,475 ✓

d. 3,400

$A(8) = 3{,}000e^{0.4} \approx 4{,}475$ dollars.

32 [Intermediate] • Target: Word problems: exponential models in context

A cup of coffee cools according to $T(t) = 22 + 70e^{-0.08t}$ (°C), where t is in minutes. Find the temperature after 15 minutes, rounded to one decimal place.

a. $38.5\,^\circ$C

b. $50.0\,^\circ$C

c. $43.1\,^\circ$C ✓

d. $22.0\,^\circ$C

$T(15) = 22 + 70e^{-1.2} \approx 43.1\,^\circ$C.

33 [Intermediate] • Target: Word problems: exponential models in context

A population of bacteria doubles every 3 hours, starting at 500. Using $P(t) = 500 \cdot 2^{t/3}$, find the population after 9 hours.

a. 4,000 ✓

b. 2,000

c. 8,000

d. 1,500

$P(9) = 500 \times 2^3 = 4{,}000$.

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34 [Intermediate] • Target: Word problems: exponential models in context

The value of a car depreciates according to $V(t) = 25{,}000(0.85)^t$ dollars, where t is in years. Find the car's value after 5 years, rounded to the nearest dollar.

a. 11,093 ✓

b. 12,750

c. 9,500

d. 10,625

$V(5) = 25{,}000(0.85)^5 \approx 11{,}093$ dollars.

35 [Intermediate] • Target: Exponentials & Logarithms

Evaluate $4^{\frac{1}{2}}$.

a. 4

b. 5

c. 2 ✓

d. 6

$4^{\frac{1}{2}} = 2$.

36 [Intermediate] • Target: Exponentials & Logarithms

In a laboratory calibration sequence, evaluate 5^2.

a. 27

b. 10

c. 23

d. 25 ✓

$5^2 = 25$.

37 [Intermediate] • Target: Exponentials & Logarithms

Solve $4^x = 256$.

a. 4 ✓

b. 3

c. 6

d. 5

$256 = 4^4$, so $x = 4$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Exponentials & Logarithms • Cross-Lesson: 1.01 Algebra & Equations

Solve $4^{x-3} = 16$.

a. 5 ✓

b. 6

c. 2

d. 4

$16 = 4^2$, so $x - 3 = 2 \Rightarrow x = 5$.

39 [Challenging] • Target: Exponentials & Logarithms • Cross-Lesson: 2.01 Functions & Transformations

Combine $\log_2 2 + \log_2 2$ into a single logarithm and evaluate.

a. 3

b. 1

c. 4

d. 2 ✓

$\log_2 (2 \times 2) = \log_2 4 = 2$ (since $2^2 = 4$).

40 [Challenging] • Target: Exponentials & Logarithms • Cross-Lesson: 3.01 Sequences & Series

Combine $\log_5 125 - \log_5 25$ into a single logarithm and evaluate.

a. 3

b. 5

c. 1 ✓

d. 2

$\log_5 \left(\frac{125}{25}\right) = \log_5 5 = 1$.

41 [Challenging] • Target: Exponentials & Logarithms • Cross-Lesson: 4.01 Trigonometry

Evaluate $\log_2 (8^2)$ using the power rule.

a. 6 ✓

b. 8

c. 5

d. 4

$\log_2 (8^2) = 2\log_2 8 = 2(3) = 6$.

42 [Challenging] • Target: Exponentials & Logarithms • Cross-Lesson: 5.01 Geometry & Coordinate Geometry

Solve $\log_5 (x - 2) = 4$.

a. 629

b. 630

c. 627 ✓

d. 625

$x - 2 = 5^4 = 625 \Rightarrow x = 627$.

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43 [Challenging] • Target: Exponentials & Logarithms • Cross-Lesson: 6.01 Limits & Derivatives

Solve $\log_5(x) = 3$.

a. 123

b. 128

c. 125 ✓

d. 127

$x = 5^3 = 125$.

44 [Challenging] • Target: Exponentials & Logarithms • Cross-Lesson: 7.01 Integration & Applications

Evaluate $\ln(e^6)$.

a. 5

b. 7

c. 6 ✓

d. e^6

$\ln(e^6) = 6$.

45 [Challenging] • Target: Exponentials & Logarithms • Cross-Lesson: 9.01 Statistics & Probability

A quantity is modeled by $N(t) = 500e^{-0.1t}$. Find $N(5)$, rounded to one decimal place.

a. 273.0

b. 500

c. 303.3 ✓

d. 333.6

$N(5) = 500e^{-0.5} = 500(0.60653) \approx 303.3$.

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